# Math 571 - Homework 2

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Problems numbered [B:2.1:3] are from Bergman's exercise notes. Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (*) to a problem, we have a slight variant of the problem.

**Problem 2.1** (R:2.2*)**.** A complex number $\gamma$ is **algebraic** iff $\gamma$ is a root to a polynomial with integer coefficients. Prove that there are complex numbers that are not algebraic.

The following are two fun facts just beyond what we got to in MAT-513:

Let $\mathbb{A} \subset \mathbb{C}$ be the set of algebraic numbers.

(a) $\mathbb{A}$ is a field.

(b) $\mathbb{A}$ is algebraically closed, that is, if $\alpha$ is a root of a polynomial in $\mathbb{A}[x]$, then $\alpha \in \mathbb{A}$. So in the definition of algebraic numbers you can use any ring of coefficients $R$, $\mathbb{Z} \subseteq R \subseteq \mathbb{A}$.

**Problem 2.2** (B:2.2:4)**.** This problem is concerning relative topology.

(a) Suppose $E$ is an open subset of a metric space $X$, and $F$ is a subset of $E$. Show that $F$ is open relative to $E$ iff $F$ is open.

(b) Suppose $E$ is a closed subset of a metric space $X$, and $F$ is a subset of $E$. Show that $F$ is closed relative to $E$ iff $F$ is open.

(c) Suppose $E$ is an open subset of a metric space $X$, and $F$ is a subset of $E$ that is closed relative to $E$. Show that $F$ need not be open or closed in $X$.

(d) Suppose $E$ is a closed subset of a metric space $X$, and $F$ is a subset of $E$ that is open relative to $E$. Show that $F$ need not be open or closed in $X$.

**Problem 2.3** (B:2.2:20)**.** Here you look at the Kuratowski game discussed in the podcast.

(a) Show that $\mathrm{Cl}(\mathrm{Cl}(E)) = \mathrm{Cl}(E)$ and $\mathrm{Int}(\mathrm{Int}(E)) = \mathrm{Int}(E)$.

(b) $\mathrm{Int}\,\mathrm{Cl}(E)) = \mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(E))))$.

(c) Show that you can get at most seven different sets from applying the operations of closure and interior repeatedly to a fixed set $E$.

(d) Show that there is $E \subset \mathbb{R}$ so that all seven possible sets are realized.

**Bonus**: If you also throw in the operation of complement, then there are at most 14 possible sets you can get starting from a fixed set $E$.

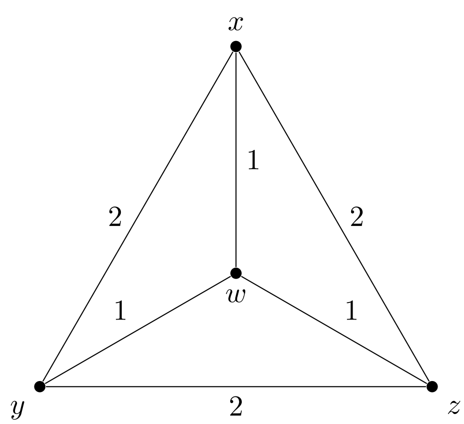
**Problem 2.4** (B:2.2:14)**.** This investigates the $p$-adic metric on $\mathbb{Z}$. Let $p > 1$ be an integer. Define $d_p(s,t) = p^{-e_p(s-t)}$ where $e_p(m) = k$ iff $p^k \mid m \wedge p^{k+1} \nmid m$, so $e_p(m)$ reports the maximal number of times $m$ can be evenly divided by $p$. Note that $p^k \mid 0$ for all $k$, so it makes sense to define $e_p(0) = \infty$ and take $d_p(s,s) = p^{-\infty} = 0$.

Show that $d_p : \mathbb{Z} \times \mathbb{Z} \to [0, \infty)$ is a metric and in fact an ***ultra-metric***, i.e., $d_p(q,s) \leq \max\{d_p(q,r), d_p(r,s)\}$.

**For Fun:** Show that all $p$-adic triangles are isosceles. This is a geometric idea; what corresponding topological fact is leading to this?

$p$-adics are very important and interesting number systems, particularly when $p$ is a prime. Here is a great video introduction to $p$-adics, which discusses applications to solving Diophantine equations.

**Problem 2.5** (B:2.2:18)**.** Let $X$ be a 4-element set. Consider a graph on the four nodes $\{x, y, z, w\}$ with edges and edge weights as shown.




(a) Verify that $d_X : X \times X \to X$ is a metric where $d(x,y) = d(x,z) = d(y,z) = 2$ and $d(w,x) = d(w,y) = d(w,z) = 1$.

(b) Show that there is no $F : X \to \mathbb{R}^k$ that is distance preserving, that is, $d_X(u,v) = d_{\mathbb{R}^k}(f(u), f(v))$.

(c) The example here is a 4-point space that cannot be *isomorphically* embedded in $\mathbb{R}^k$ for any $k$, yet every 3-point subspace can be isomorphically embedded in $\mathbb{R}^k$. Can you find a 5-point space that cannot be isomorphically embedded in $\mathbb{R}^k$ and yet all of whose 4-point subspaces can be isomorphically embedded in $\mathbb{R}^k$ for sufficiently large $k$.

**Problem 2.6** (B:2.2:19)**.** Let $X$ be an infinite metric space. Show that there are uncountably many open subsets of $X$.

(a) Show that you can find $p_i \in X$ and $r_i > 0$ so that $N_{r_i}(p_i) \cap N_{r_j}(p_j) = \emptyset$.

(b) Associate with every $\alpha \in 2^{\mathbb{N}}$ the open set $O_x = \bigcup_{\{i \mid x(i)=1\}} N_{r_i}(p_i)$. Show that this map is an injection of $2^{\mathbb{N}}$ into the open sets of $X$.

(c) (b) shows that there are at least continuum many open subsets of $X$, could the cardinality be even larger?

**Problem 2.7** (R:2.26)**.** Let $X$ be a metric space in which every infinite set, $A$, has a limit point in $A$. Show that $X$ is compact. (Follow the hints in Rudin.)

**This is a very important property.** We will see that for metric spaces, this gives the equivalence of "$X$ compact" with "$X$ is sequentially compact."

**Problem 2.8** (B:2.3:3 - Compact sets get crowded!)**.** Let $X$ be a compact metric space and let $\epsilon > 0$. Show that there is an $N$ so that for any $m \geq N$, if $p_1, \ldots, p_m$ are $m$ distinct points in $X$, then at least two of them are within $\epsilon$ of each other, that is, there are $i, j$ with $d(p_i, p_j) < \epsilon$.

**Problem 2.9** (B:2.5:1 - The usual definition of connected.)**.** (a) Show that $E \subsetneq X$ is connected (as per Rudin) iff there is $\emptyset \neq A \subsetneq E$ that is ***clopen***, i.e., closed and open.

(b) Show that $E \subsetneq X$ is connected (as per Rudin) iff there are two non-empty ***relatively open subsets*** of $E$ that ***cover*** $E$.

**Note:** A space $X$ is connected iff the clopen sets consist of exactly $X$ and $\emptyset$.

**Problem 2.10** (This is a mix of problems.)**.** Consider the following collections of open subsets of $\mathbb{Q}$. (1) Verify that each gives a topology on $\mathbb{Q}$ and (2) compute for each, $\mathrm{Cl}((0, \sqrt{2}))$, (3) show that the space is compact, or not, (4) show that the space is connected or not, and finally (5) how are these related to the standard relative topology that $\mathbb{Q}$ inherits from $\mathbb{R}$?

(a) *The left-ray topology.* Take $\mathcal{O}$ to be the set of all right-rays of the form $(q, \infty)$ for $q \in \mathbb{Q} \cup \{\infty\}$.

(b) *The co-finite topology.* Take $\mathcal{O}$ to be subsets of $\mathbb{Q}$ whose complements are finite.

(c) *The order topology.* A set $A$ is open in this space iff for all $q \in A$, there is $q_1 < q < q_2$ in $\mathbb{Q}$ so that $(q_1, q_2) \cap Q \subseteq A$.